BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 262 · LMC instructions and data

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does direct addressing [30] retrieve?

Q2. What is the accumulator?

Q3. What is a mnemonic?

Q4. Search for 19 in sorted [1,4,7,10,13,16,19,22,25]. Show midpoint indexes checked.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LMC — Little Man Computer teaching instruction set.

ACC — Accumulator holding current working value.

mailbox — Numbered LMC memory location.

STA — Store accumulator to memory.

LDA — Load accumulator from memory.

DAT — Declare/initialise a data location.

New learning

Know LMC's parts — LMC uses numbered mailboxes to hold instructions and data, and an accumulator for working values.

Trace instructions in order — First INP enters 4 into ACC, then STA FIRST copies 4 to the data mailbox before another input overwrites ACC.

Produce the result — Second INP makes ACC=7; ADD FIRST adds stored 4 to make 11, and OUT prints 11. HLT ends the program.

Worked example

Write and trace an LMC program adding inputs 3 and 5.

1. Predict — Store the first input 3 into NUM so the second input will not overwrite it.

2. Explain — Read the second input 5, add NUM to get 8 and output the result.

We Do · Practise with guidance

Write and trace LMC code to calculate first input 9 minus second input 4.

```
INP
STA A
INP
STA B
LDA A
SUB B
OUT
HLT
A DAT 0
B DAT 0
```

Predict

Store input 9 as A and input 4 as B, since the accumulator is overwritten by INP.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Explain

LDA A restores the first input 9 and SUB B produces 5; OUT prints 5.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain what INP, OUT, LDA and STA do in LMC.

Question 2

Trace INP; STA X; INP; ADD X; OUT; HLT; X DAT 0 using inputs 6 and 8. State each change and output.

Question 3

Write an OCR LMC program that takes two inputs from 0 to 999, with the first greater than or equal to the second. Subtract the second from the first and output the non-negative result. Trace inputs 9 and 2.

Exit Ticket · Answer before checking

In LMC, STA X replaces ACC with the stored value of X.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The data at memory address 30.

Q2. A CPU register holding an arithmetic/intermediate value.

Q3. A readable abbreviation of an operation code.

Q4. low0 high8 mid4→13, low5 high8 mid6→19; found index 6.

You Do pages · Independent questions

Q1. [4] INP takes input to ACC [1]; OUT outputs ACC [1]; LDA loads ACC from mailbox [1]; STA stores ACC to mailbox [1].

Q2. [6] First INP makes ACC6 [1]; STA X stores X6 [1]; second INP makes ACC8 [1]; ADD X makes ACC14 [2]; OUT prints 14 [1].

Q3. [6] INP; STA A [1]; INP; STA B [1]; LDA A [1]; SUB B [1]; OUT; HLT and A DAT 0; B DAT 0 [1]; output 7 [1]. Assume inputs and the non-negative result are representable in LMC's three-digit mailboxes.

Exit Ticket · Model answer

False. STA X copies ACC into memory X; LDA X loads the content of X into ACC.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.